THE SPECIAL FIBRE OF A MODIFICATION OR ALTERATION OVER A LOCAL DOMAIN HAS DIMENSION AT MOST DIM(A) - 1

THE STACKS PROJECT AUTHORS

Editorial note. The statement carries a source placeholder '(5) add more here.'; recorded verbatim in the span, harmless since the proof handles (1)-(4) through (4).

Allowed standard prerequisites (editorial).

- Generic points of a decent space under an etale cover. Decent Spaces, lemma decent-generic-points (cross-chapter, named). Identifies generic points of the affine etale cover U = Spec(B) with those of |X|.
- Generically finite morphisms. Algebra, lemma generically-finite (cross-chapter, named). Produces a nonzero f in A with A_f -> B_f finite after localizing at the minimal primes.
- Dimension formula for finitely generated algebras. Algebra, lemma dimension-formula (cross-chapter, named). Bounds dim(B_q) by dim(A) for a prime q over m.

Source reference. Original reference 02IJ: focused-seeds/geometry/S9-spres-dimension-special-fibre/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02ML: focused-seeds/geometry/S9-spres-dimension-special-fibre/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0ABV: focused-seeds/geometry/S9-spres-dimension-special-fibre/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04MU: focused-seeds/geometry/S9-spres-dimension-special-fibre/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-resolve.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0ADA: focused-seeds/geometry/S9-spres-dimension-special-fibre/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0AD8: focused-seeds/geometry/S9-spres-dimension-special-fibre/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $(A, \mathfrak{m}, \kappa)$ be a Noetherian local domain of dimension ≥ 1. Let $f : X \to \mathrm{Spec}(A)$ be a morphism of algebraic spaces. Assume at least one of the following conditions is satisfied*

1. *f is a modification (Spaces over Fields, Definition 0AD8),*
2. *f is an alteration (Spaces over Fields, Definition 0ADA),*
3. *f is locally of finite type, quasi-separated, X is integral, and there is exactly one point of $|X|$ mapping to the generic point of $\mathrm{Spec}(A)$,*
4. *f is locally of finite type, X is decent, and the points of $|X|$ mapping to the generic point of $\mathrm{Spec}(A)$ are the generic points of irreducible components of $|X|$,*
5. *add more here.*

Then $\dim(X_\kappa) \leq \dim(A) - 1$.

Proof. Cases (1), (2), and (3) are special cases of (4). Choose an affine scheme $U = \mathrm{Spec}(B)$ and an étale morphism $U \to X$. The ring map $A \to B$ is of finite type. We have to show that $\dim(U_\kappa) \leq \dim(A) - 1$. Since X is decent, the generic points of irreducible components of U are the points lying over generic points of irreducible components of $|X|$, see Decent Spaces, Lemma 0ABV. Hence the fibre of $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ over (0) is the (finite) set of minimal primes $\mathfrak{q}_1, \ldots, \mathfrak{q}_r$ of B. Thus $A_f \to B_f$ is finite for some nonzero $f \in A$ (Algebra, Lemma 02ML). We conclude $\kappa(\mathfrak{q}_i)$ is a finite extension of the fraction field of A. Let $\mathfrak{q} \subset B$ be a prime lying over $\mathfrak{m}$. Then

$$\dim(B_\mathfrak{q}) = \max \dim((B/\mathfrak{q}_i)_\mathfrak{q}) \leq \dim(A)$$

the inequality by the dimension formula for $A \subset B/\mathfrak{q}_i$, see Algebra, Lemma 02IJ. However, the dimension of $B_\mathfrak{q}/\mathfrak{m}B_\mathfrak{q}$ (which is the local ring of U_κ at the corresponding point) is at least one less because the minimal primes $\mathfrak{q}_i$ are not in $V(\mathfrak{m})$. We conclude by Properties, Lemma 04MU. □